Protein Engineering

Module: SC3282

By

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Lecture 05:

Structure-functional relationships for protein design

Levels of Protein Structure

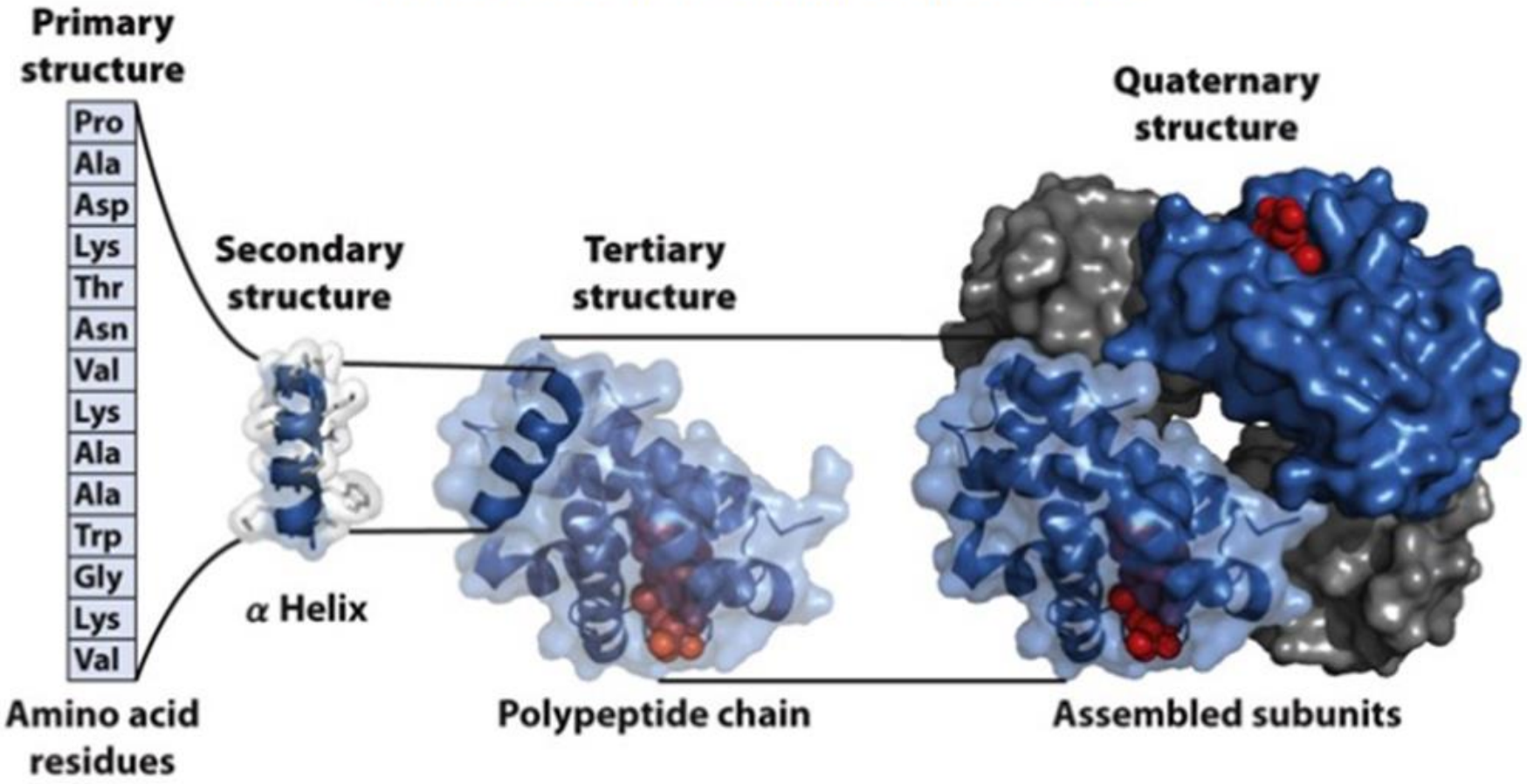

Figure 3-23
Lehninger Principles of Biochemistry, Fifth Edition
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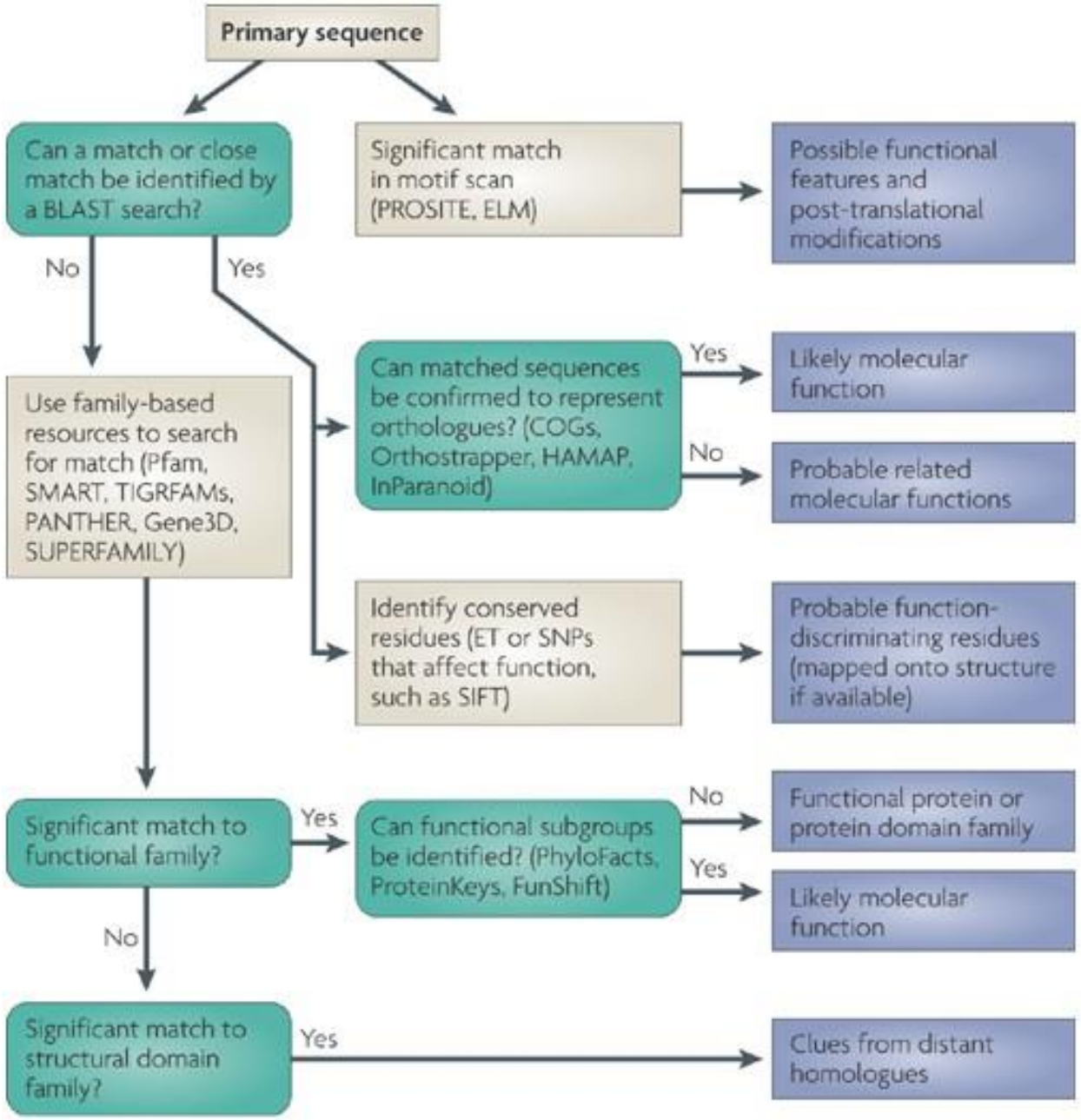
Predicting protein function from sequence and structure



Nature Reviews | Molecular Cell Biology

Methods determine the structure of a protein:

- Several methods are currently used to determine the structure of a protein,
    - X-ray crystallography,
    - NMR spectroscopy, and
    - Electron microscopy.

- Each method has advantages and disadvantages. In each of these methods, the scientist uses many pieces of information to create the final atomic model.

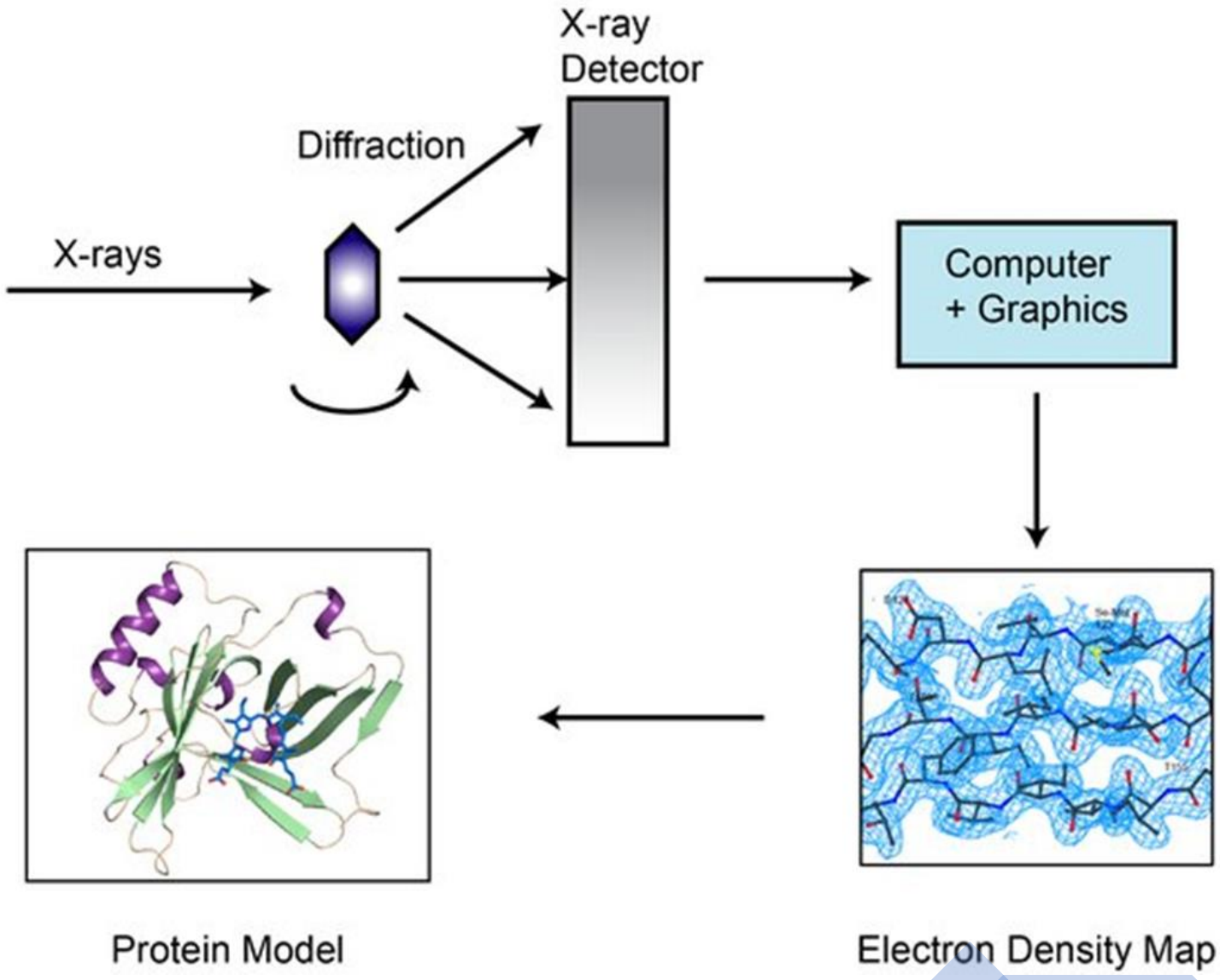
X-ray crystallography

- For this method, the protein is purified and crystallized, then subjected to an intense beam of X-rays.
- The proteins in the crystal diffract the X-ray beam into one or another characteristic pattern of spots, which are then analyzed (with some tricky methods to determine the phase of the X-ray wave in each spot) to determine the distribution of electrons in the protein.
- The resulting map of the electron density is then interpreted to determine the location of each atom.
- The PDB archive contains two types of data for crystal structures. The coordinate files include atomic positions for the final model of the structure, and the data files include the structure factors (the intensity and phase of the X-ray spots in the diffraction pattern) from the structure determination.

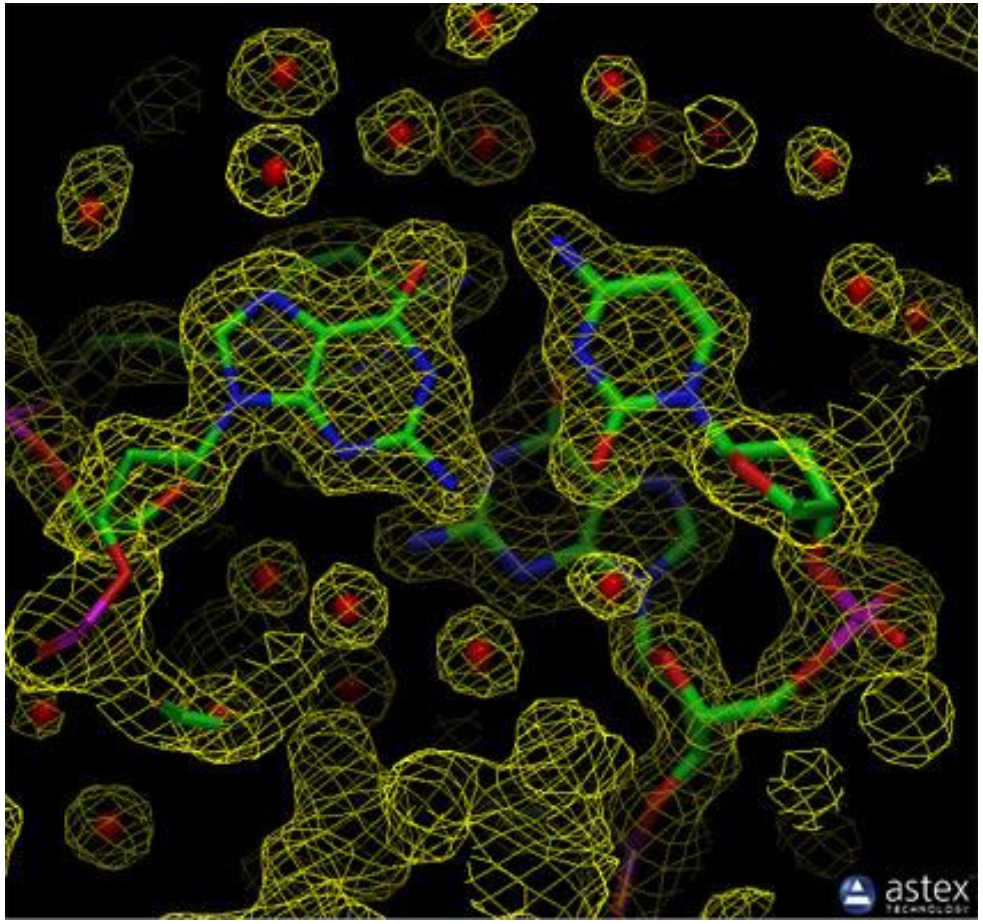
X-ray crystallography

- Can provide very detailed atomic information, showing every atom in a protein or nucleic acid along with atomic details of ligands, inhibitors, ions, and other molecules that are incorporated into the crystal.
- However, the process of crystallization is difficult and can impose limitations on the types of proteins that may be studied by this method.
- For example, X-ray crystallography is an excellent method for determining the structures of rigid proteins that form nice, ordered crystals.
- Flexible proteins, on the other hand, are far more difficult to study by this method because crystallography relies on having many, many molecules aligned in exactly the same orientation, like a repeated pattern in wallpaper.
- Flexible portions of protein will often be invisible in crystallographic electron density maps, since their electron density will be smeared over a large space.

X-ray crystallography



The experimental electron density from a structure of DNA is shown here (PDB entry 196d), along with the atomic model that was generated based on the data. The contours surround regions with high densities of electrons, which correspond to the atoms in the molecule.



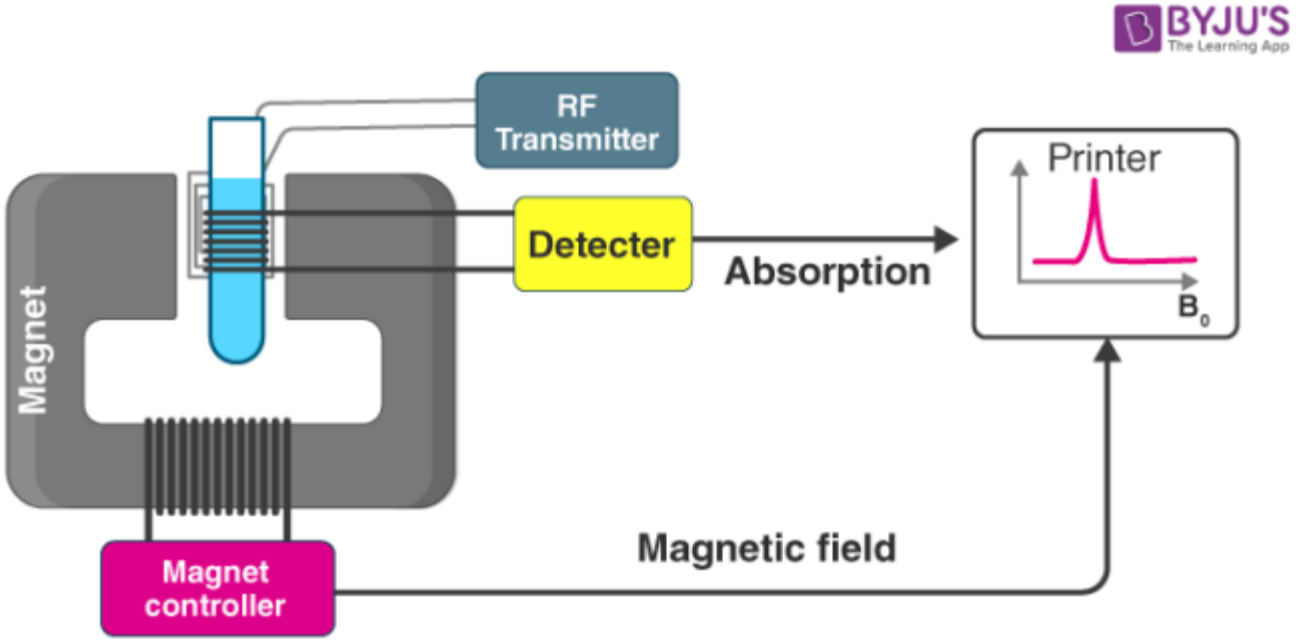
X-ray crystallography-

https://www.youtube.com/watch?v=8SN1kmfhWcs

https://www.youtube.com/watch?v=RUok4O9oovQ

https://www.youtube.com/watch?v=xBA09PXPPR4

NMR spectroscopy



NMR Spectroscopy Instrumentation

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NMR spectroscopy may be used to determine the structure of proteins.

- The protein is purified, placed in a strong magnetic field, and then probed with radio waves.
- A distinctive set of observed resonances may be analyzed to give a list of atomic nuclei that are close to one another, and to characterize the local conformation of atoms that are bonded together.
- This list of restraints is then used to build a model of the protein that shows the location of each atom.

NMR spectroscopy

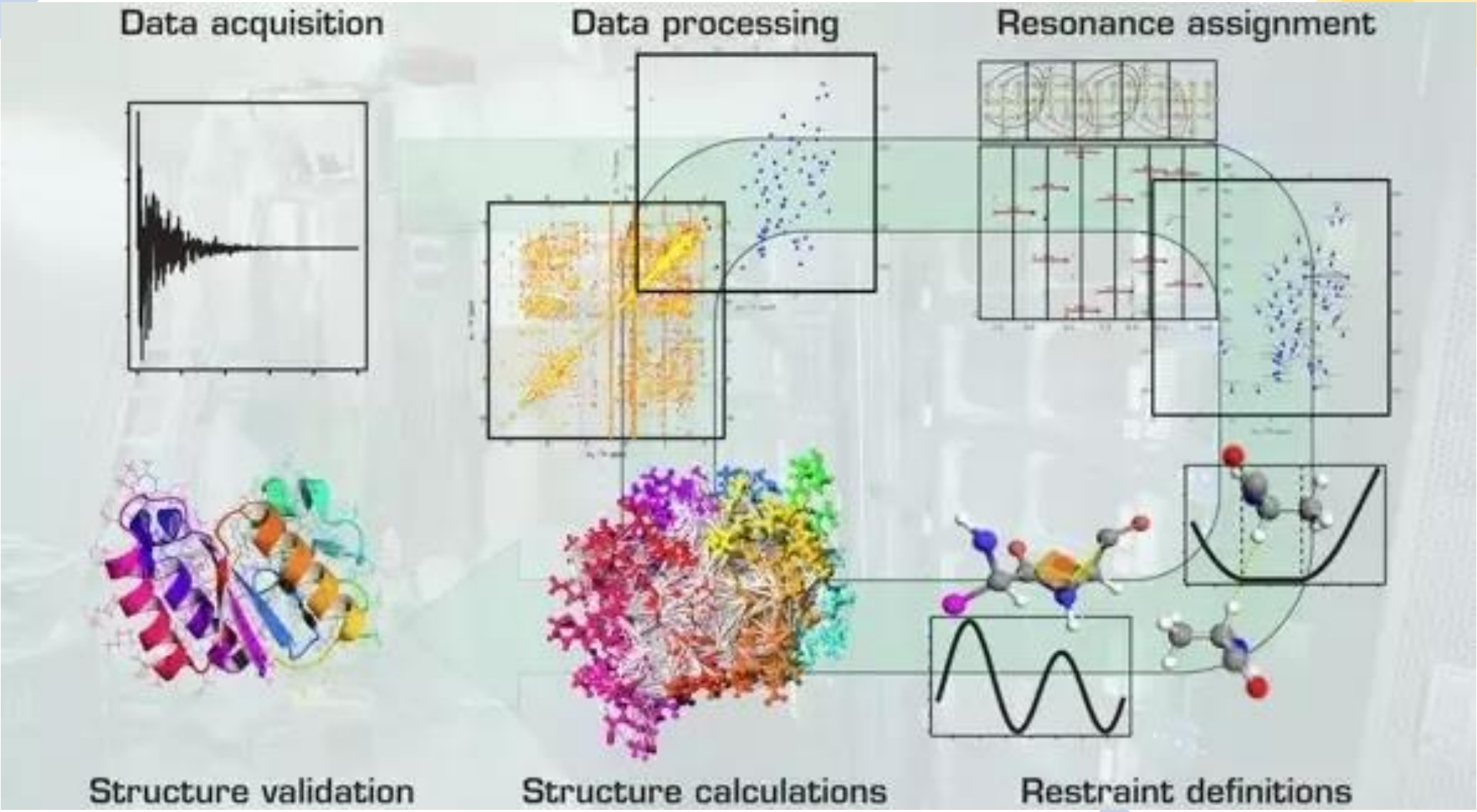
- The technique is currently limited to small or medium proteins, since large proteins present problems with overlapping peaks in the NMR spectra.

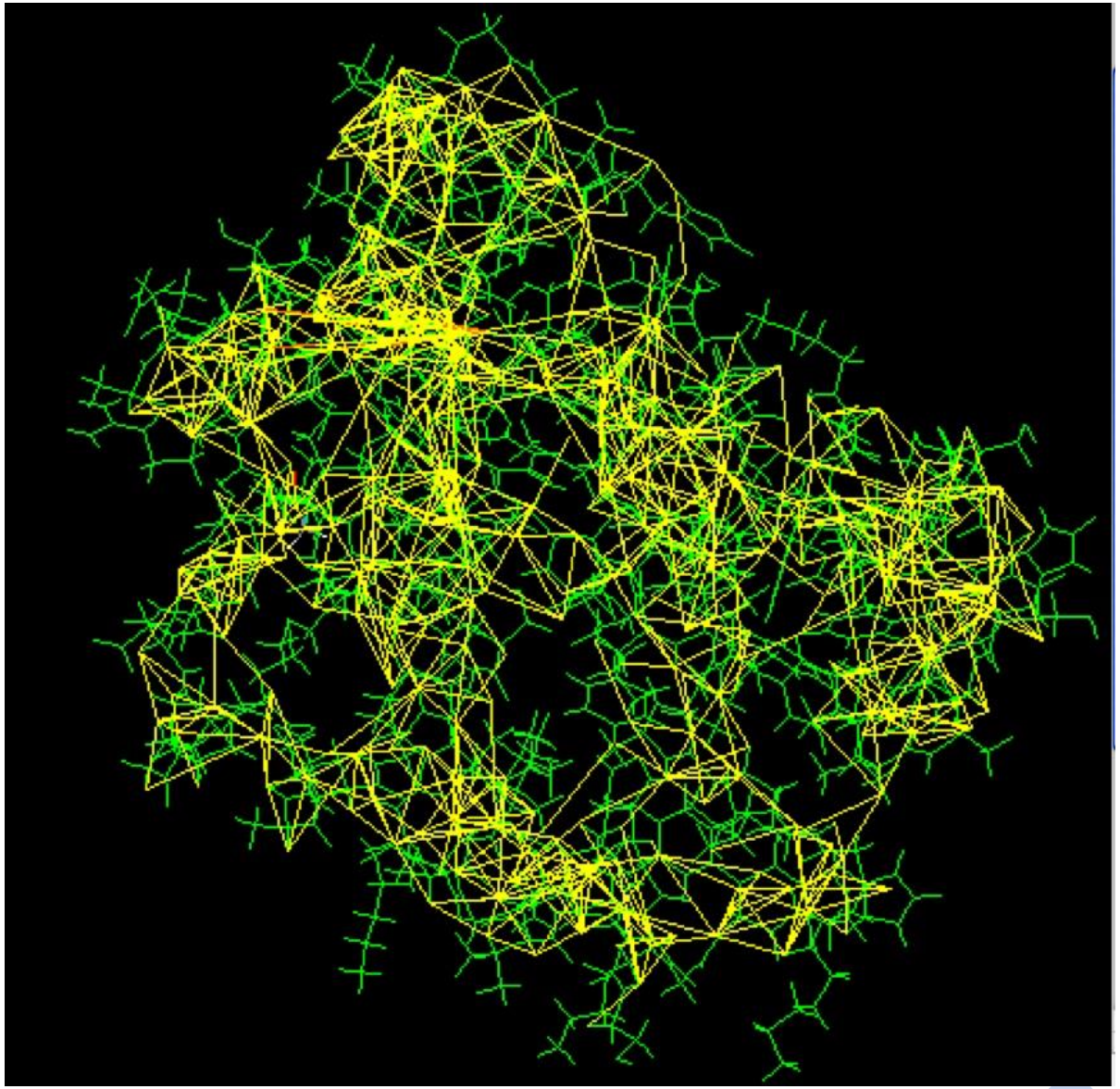
- A major advantage of NMR spectroscopy is that it provides information on proteins in solution, as opposed to those locked in a crystal or bound to a microscope grid, and thus, NMR spectroscopy is the premier method for studying the atomic structures of flexible proteins.

- A typical NMR structure will include an ensemble of protein structures, all of which are consistent with the observed list of experimental restraints.

- The structures in this ensemble will be very similar to each other in regions with strong restraints, and very different in less constrained portions of the chain. Presumably, these areas with fewer restraints are the flexible parts of the molecule, and thus do not give a strong signal in the experiment.

NMR-
https://www.youtube.com/watch?v=t8JIBwLsCGA&t=54s


Data acquisition
Data processing
Resonance assignment
Structure validation
Structure calculations
Restraint definitions



3D Electron Microscopy:

- Electron microscopy, frequently referred to as 3DEM, is also used to determine 3D structures of large macromolecular assemblies.

- A beam of electrons and a system of electron lenses is used to image the biomolecule directly. Several tricks are required to obtain a 3D structure from 2D projection images produced by transmission electron microscopes.

- The most commonly used technique today involves imaging of many thousands of different single particles preserved in a thin layer of non-crystalline ice (cryo-EM).

3D Electron Microscopy:

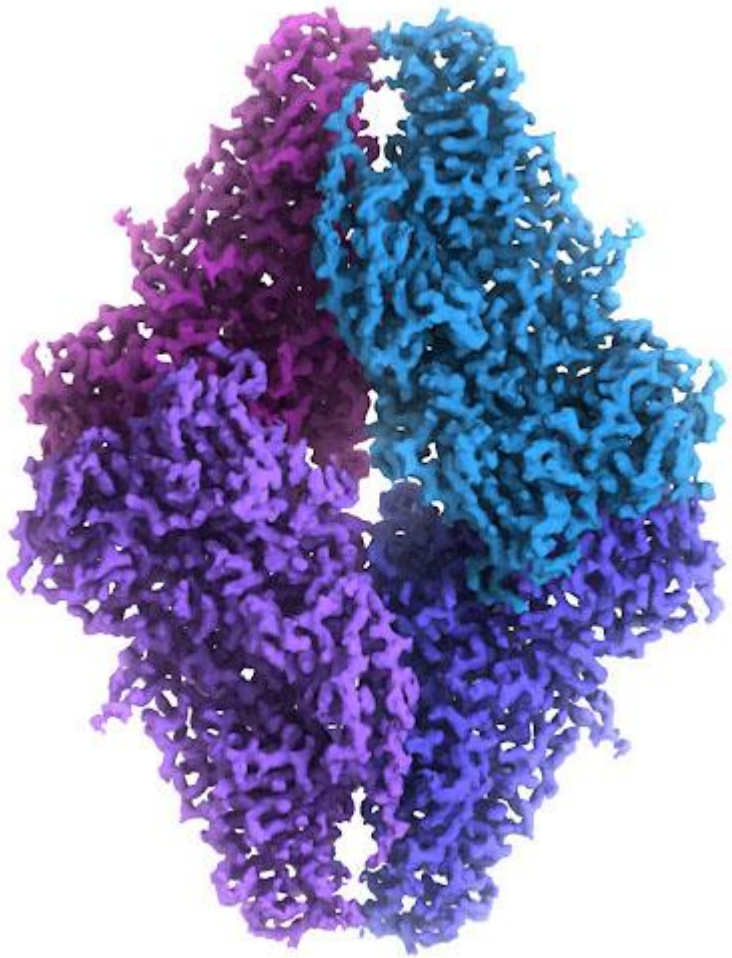
- With a sufficient number of single particles, the 3DEM maps can then be interpreted by fitting an atomic model of the macromolecule into the map, just as macromolecular crystallographers interpret their electron density maps.

- In a restricted number of cases, electron diffraction from 2D or 3D crystals or helical assemblies of biomolecules can be used determine 3D structures with an electron microscope using an approach very similar to that of X-ray crystallography.

beta-galactosidase:

- This cryo-EM map of beta-galactosidase was built from over 90,000 images of the molecule frozen in ice, which was detailed enough to provide an atomic model. The cryoEM map is at EMDataBank entry EMD-2984, and the atomic coordinates are in PDB entry 5a1a.

- Image courtesy of Veronica Falconieri and Siriam Subramaniam, National Cancer Institute.



Electron microscopy-
https://www.youtube.com/watch?v=vLo7oqfRa74&t=57s

Integrative Modeling:

- Researchers are interested in studying larger and more complex systems, and use every technique available to do so.

- The structural biology community has had particular success in recent years by using an approach, termed “integrative modeling.”

- The idea is to combine information from a variety of methods, each good for studying a particular aspect of the system, to create an overall picture of the assembly.

- For instance, combining spectroscopic or chemical crosslinking data that identify distances between components in an assembly, with low resolution electron microscopy data that give information on the overall shape of a complex, has become an effective strategy in integrative modeling.

Integrative Modeling:

- In addition to traditional structural biology methods such as X-ray crystallography, NMR spectroscopy and electron microscopy, experimental methods such as small angle solution scattering, Forster resonance energy transfer, chemical crosslinking, mass spectrometry, electron paramagnetic resonance spectroscopy, and other biophysical techniques have been used in integrative modeling studies.

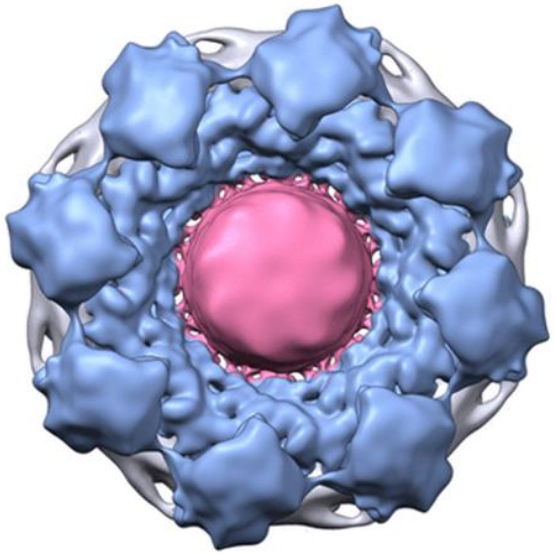
- A key aspect of integrative modeling is that the resulting structural models do not always comprise of atomic coordinates and can contain regions of coarse-grained beads that represent multiple atoms.

- This is due to the fact that different kinds of experiments provide information at different levels of resolution.

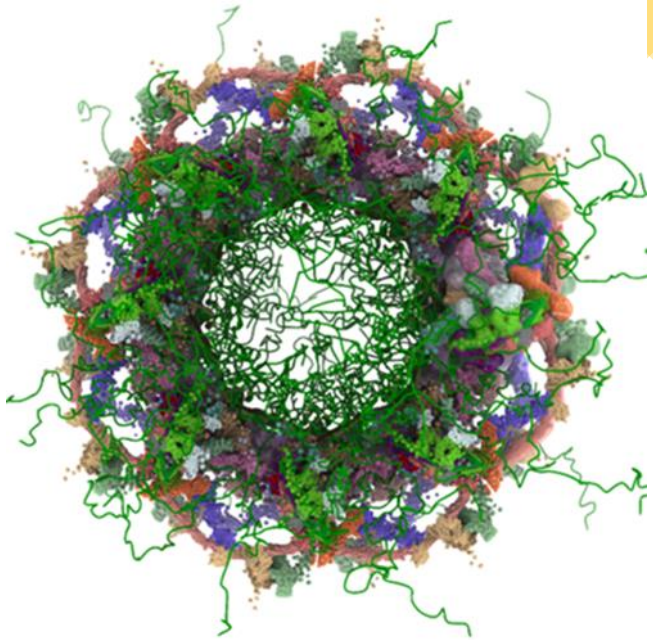
Integrative Modeling:

- An example of integrative modeling is the structure of the nuclear pore complex (NPC) from budding yeast determined using data from chemical crosslinking, small angle solution scattering and electron microscopy experiments.

- The NPC is an eight-fold symmetric assembly consisting of 552 copies of 32 different proteins belonging to the nucleoporin family.

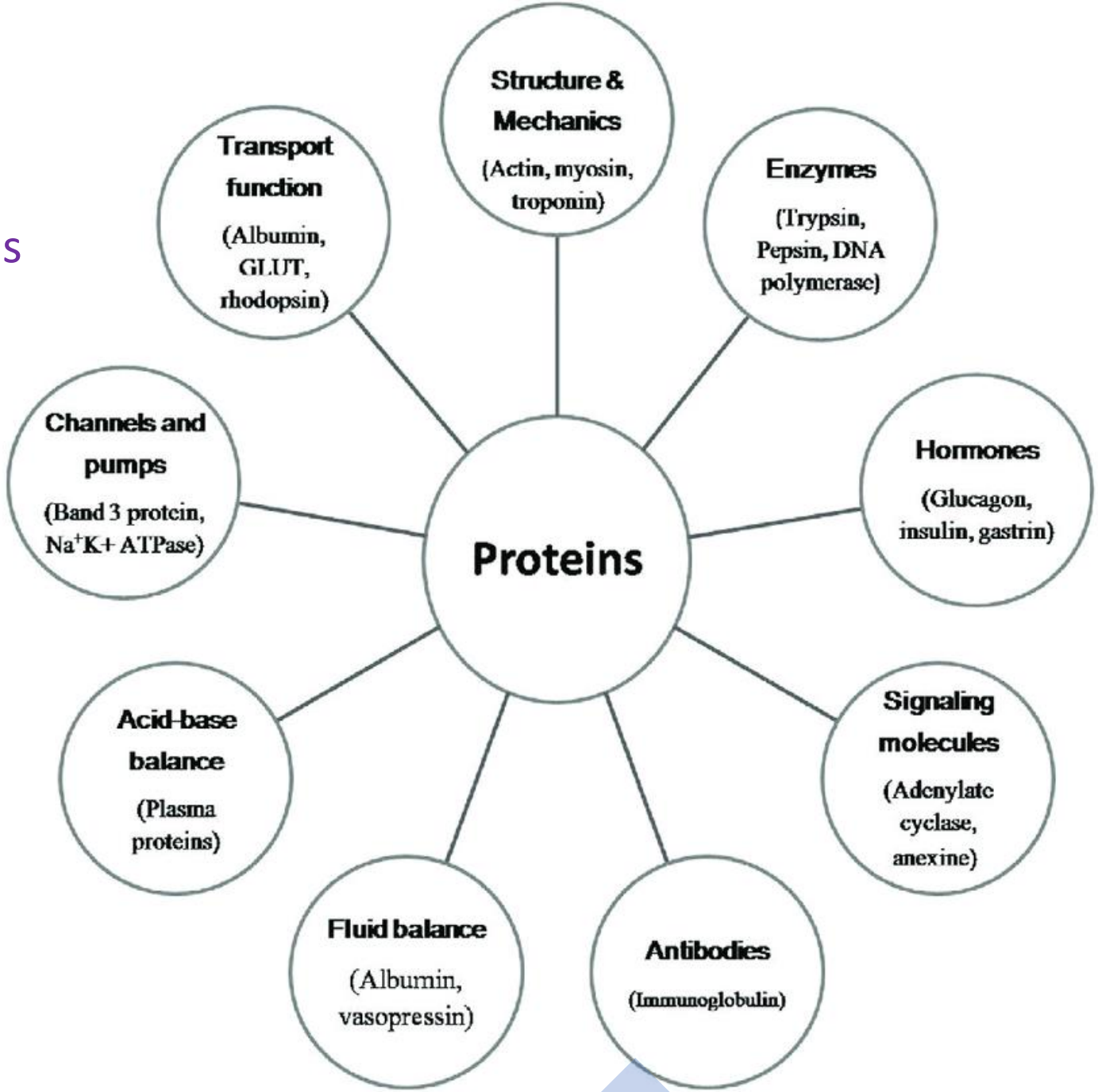


Cryo electron microscopy map of the NPC from EMD-7321



The integrative model of the NPC from PDBDEV_00000012.

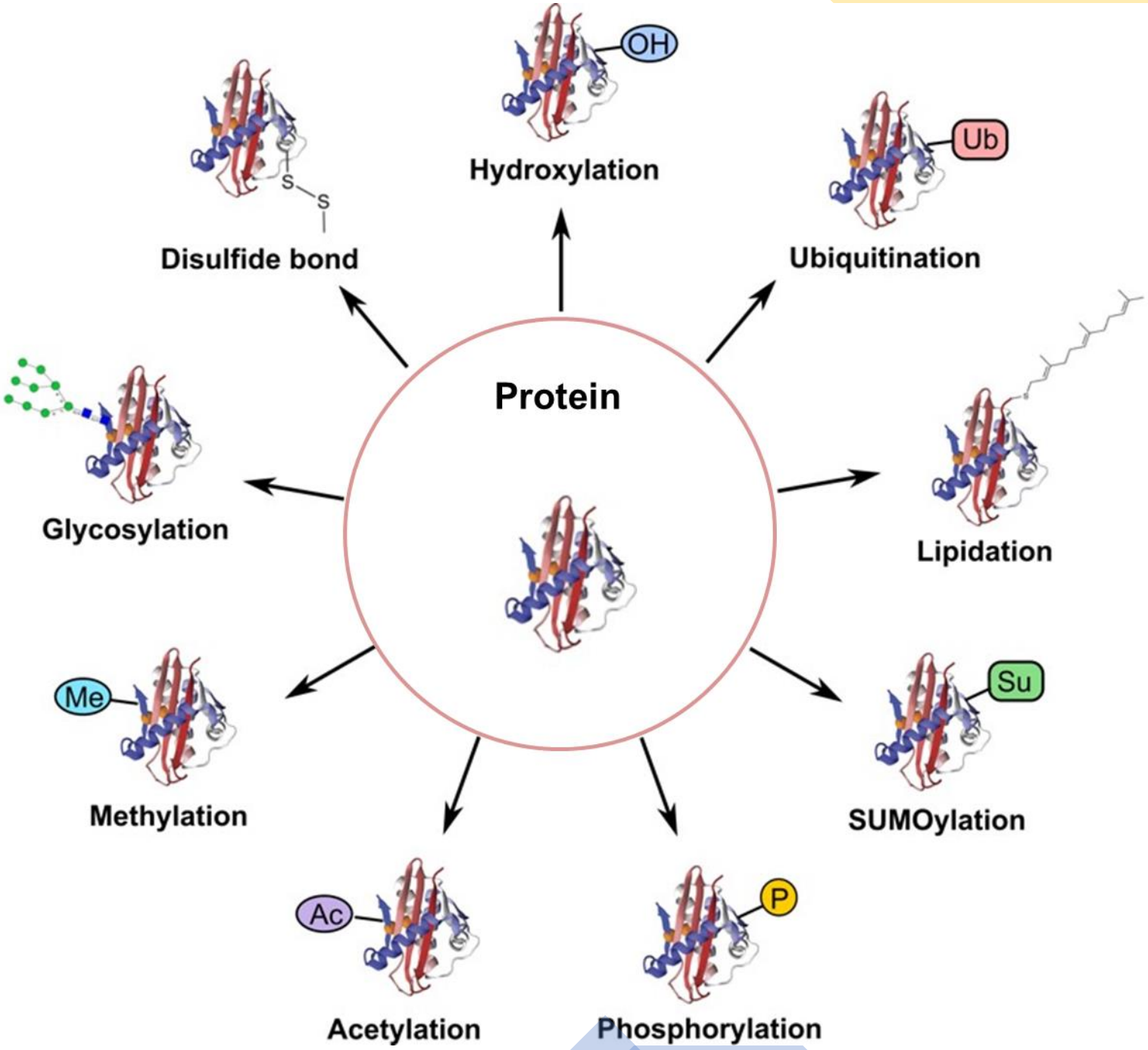
Multiple functions of proteins



Posttranslational Modifications of Proteins:

- The human genome contains 23 000 genes. mRNA-processing (alternative splicing, mRNA editing etc.) results in 3 mRNAs per gene.
- Post-translational modification of the proteins produced from them creates 10 different protein species from each mRNA.
- Thus, the human proteome consists of 10^6 proteins, with different functions, regulation, destruction.
- The properties of proteins can be changed by post-translational modification; in some cases, this can be done (or undone) quickly in response to environmental stimuli, for example, exposure to hormones.
- Such modifications can switch enzymes between active and inactive states and are required for the proper targeting of a protein to subcellular structures. The following reactions are of particular importance:

Post-translational Modifications of Proteins



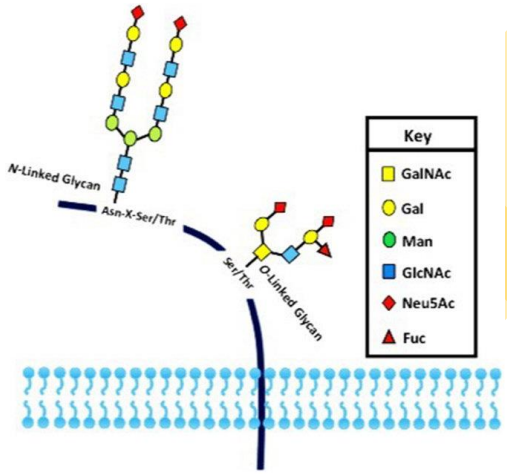
Glycosylation

- Glycosylation is the process of enzymatic transfer of oligosaccharide (sugar) trees to proteins.

- They are affixed either to the OH-groups of Ser or Thr (O-linked) or to the acid amide group of Asn (N-linked).

- Other amino acids (Arg, Tyr, Trp, Hyl, Hyp) are involved much less frequently, e.g., in collagen. Addition occurs in the ER and the golgi-apparatus to the extracellular domain of membrane proteins and to secreted proteins.

- Cytosolic proteins are rarely glycosylated. In bacteria, glycosylation occurs in the periplasm.



Glycosylation:

- Glycosylation is required for proper protein folding. Glycosylation inhibitors are used as antiviral drugs (e.g. nojirimycin or desoxynojirimycin).
- Sugar trees are also required as “address labels” in the intracellular transport of proteins between compartments.
- On the cell surface, the sugar trees of membrane proteins serve as recognition sites for cell–cell-interactions, as immunological determinants and—since everything has to have a downside too—as docking sites for bacteria and virus.

Glycation :

- In glycosylation sugars are enzymatically transferred to proteins in a carefully orchestrated process.

- Glycation (sometimes called non-enzymatic glycosylation) is the covalent attachment of a sugar to a protein or lipid. Typical sugars that participate in glycation are glucose, fructose, and their derivatives.

- Glycation, however, is a spontaneous reaction between sugar aldehydes (and some other reducing compounds) and amino groups in proteins, nucleic acids, and lipids.

- The velocity of Glycation depends on the concentration of glucose in the blood. This has a direct medical application: In diabetics, the concentration of Glycation hæmoglobin (HbA1c) depends on the average blood glucose concentration during the lifespan of an erythrocyte (about 3months).

Disulfide Bond Formation:

- Oxidation of the SH-groups of two cysteine residues leads to the formation of a covalent bond. The cell uses the tripeptide glutathione as reducing agent.
- Disulfide bond formation does not happen in the reducing environment of the cytosol, but in the ER (or the bacterial periplasm) which is oxidizing.
- Special enzymes, Protein Disulfide isomerase (PDI), make sure that the right Cys residues undergo Disulfide bond formation.

Proteolysis:

- Proteolysis is involved in the activation of proenzymes, for example, in the digestive system.
- Digestive enzymes (e.g. trypsin, chymotrypsin) are produced as inactive precursors (zymogens), so that they cannot harm the cells secreting them.
- Once released into the intestine, they are activated by cleaving off a part of the enzyme that was blocking the active site.
- Cascades of proteolytic enzymes make up our blood-clotting and complement system.
- Prohormones (e.g. insulin) are activated in a similar manner. On the other hand, proteins no longer needed can be inactivated by proteolysis (e.g., cyclins in cell cycle).
- Proteolysis may also be used to remove signal peptides. For example, some proteins destined for the intermembrane space of mitochondria carry a signal sequence for mitochondrial import which leads to their import into the mitochondrial matrix.
- There the signal peptide is cleaved off by matrix protease, exposing a second signal directing the protein's export into the intermembrane space through a different transporter.

Hydroxylation:

- Protein hydroxylation occurs on Pro and Lys residues.
- Proteins regulated by Pro-hydroxylation are the hypoxia induced transcription factors (HIF).
- As a consequence, oxygen consumption of the cell is down-regulated; it can survive a low oxygen supply for a longer time.

Phosphorylation/Dephosphorylation:

- The transfer of phosphoryl groups from ATP to the hydroxy groups of Ser, Thr, and Tyr (rarely onto His-nitrogen or Asp and Glu COO-) is important for the reversible regulation of enzyme activity. The transfer is catalysed by protein kinases, and the removal by protein phosphatases.

- Thus, the reaction is rapidly reversible at minimal expense for the cell (a single high energy phosphate bond).

- One-third of all proteins in the cell undergo regulatory phosphorylation/ dephosphorylation cycles.

Acetylation/Deacetylation:

- Transfer of acetyl groups from acetyl-CoA onto the –-amino group of Lys by protein acetylases, and their removal by protein deacetylases, are also used for regulation of enzymatic activity.

- The human acetylome, containing 1750 proteins, has recently been determined.

- Many DNA-binding proteins are regulated by acetylation, because the acetylated Lys is much less likely to be protonated, hence less likely to bind to the negative charges on DNA.

- In addition, the change in protonation also affects the binding of transcription factors. The activity of metabolic enzymes may also be regulated by Lys-acetylation, for example, the glycolytic activity of Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) is increased by acetylation; gluconeogensis is stimulated by deacetylation.

Methylation/Demethylation:

- Transfer of methyl groups from S-adenosyl methionine (SAM) onto proteins may also serve regulatory purposes, but we know very little about it.

- Transfer can be to carboxyl groups, forming methyl esters. This reaction is used to mark damaged proteins for destruction, but also in signal cascades of unknown function.

Addition/Removal of Hydrophobic Tails:

- Addition of
  palmitoyl- (fatty acid) groups to internal Cys or Ser
  myristoyl- (fatty acid) groups to N-terminal Gly
  farnesyl- or geranylgeranyl (isoprenoid) groups to C-terminal Cys, converts cytosolic enzymes to membrane-bound (cytosolic leaflet).

- Because this is required for the activation of some enzymes, the transferases make possible drug targets (e.g., anticancer drugs).

S-Nitrosylation:

- S-Nitrosylation occurs on Cys-residues.
- The Cys must be positioned between a basic and an acidic residue (either in primary, tertiary, or quaternary structure) because of acid-base catalysis.
- Nitrosylation serves as an additional pathway for NO regulation besides cGMP dependent kinases.

ADP-Ribosylation: Addition of ADP-ribose moieties to a protein.

- ADP-Ribosylation is used by some bacterial toxins (Vibrio cholerae, Bordetella pertussis) to inactivate cellular proteins.
- This is the starting point of the patho-mechanism of the diseases associated with these bacteria (cholera and whooping cough, respectively).

Deamidation:

- Deamidation is the removal of the acid amide group from Gln or Asn, forming Glu and Asp, respectively. It may be followed by racemization (formation of D-amino acids).

AMPylation (Adenylylation):

- AMPylation (Adenylylation) is performed from ATP onto critical Thr hydroxy groups of Rho GTPases and other important regulatory proteins by Vibrio parahaemolyticus Vibrio outer protein S (VopS).

- This results in depolymerization of the actin cytoskeleton, affected cells round up.

Transfer of Peptides:

- Ubiquitin is a highly conserved 76-amino-acid protein, which is ubiquitously expressed in all tissues in eukaryotes, whose major function is to selectively label proteins for degradation.

Thank you